Exercises from Lectures 1–3. Throughout, $\Bbbk$ is a field, and all modules are right modules unless a left action is specified. Write $\Bbbk A_n$ for the algebra of upper triangular $n \times n$ matrices over $\Bbbk$, with matrix units e_{ij} $(1 \le i \le j \le n)$ and grading $\deg e_{ij} = j - i$. For a graded module M, the shift convention is $M(r)_d = M_{d+r}$. Ideals in algebra presentations are two-sided ideals.

Lecture 1: Graded algebras, modules, and duals

Exercise 1. Graded modules and homogeneous maps
The following parts are independent.

(1) Let $n \ge 1$ and $A = \Bbbk A_n$. Verify that

$$A_d = \operatorname{span}_{\Bbbk}\{e_{ij} \mid j - i = d\}$$

is a grading, using $e_{ij}e_{pq} = \delta_{jp}e_{iq}$. Compute $\operatorname{rad}(A)$ from its definition as the intersection of all maximal right ideals.

(2) Let $A = \Bbbk A_3$ and $M = e_{22}A$. Its graded linear dual is

$$\mathbb{D}(M)_d = \operatorname{Hom}_{\Bbbk}(M_{-d}, \Bbbk), \qquad (af)(m) = f(ma) \quad (a \in A,\ f \in \mathbb{D}(M),\ m \in M).$$

Find a homogeneous basis of $\mathbb{D}(M)$, its degrees, and all non-zero actions of the matrix units on that basis. Show that $\mathbb{D}(M)$ is not isomorphic to any left module $(Ae_{jj})(r)$, for $j \in \{1, 2, 3\}$ and $r \in \mathbb{Z}$.

Hint. Compare dimensions and the actions of e_{11}, e_{22}, e_{33}.

(3) Let $S = \Bbbk e_1 \oplus \Bbbk e_2$, with $e_ie_j = \delta_{ij}e_i$ and $1 = e_1 + e_2$, in degree 0. Set

$$M = \bigoplus_{i \ge 0} \Bbbk m_i, \qquad \deg m_i = i, \quad m_ie_1 = m_i, \quad m_ie_2 = 0,$$

and let $N = \Bbbk e_1$ be in degree 0. Define the S-linear map $f : M \to N$ by $f(m_i) = e_1$ for every $i \ge 0$. Write every non-zero homogeneous component $f_d : M \to N$ explicitly, where a degree d map sends M_j into N_{j+d}. Prove that f is not a finite sum of homogeneous maps.

(4) Let $A = \Bbbk A_3$ and $P_i = e_{ii}A$ for $i \in \{1, 2, 3\}$, with the inherited grading.

i. For each of the nine ordered pairs (i, j), find a basis of $\operatorname{Hom}_A(P_i, P_j)$ and the degree of every basis map. Present the answers in a 3×3 table whose rows index the source P_i and whose columns index the target P_j.

ii. Compute the composite

$$P_3 \xrightarrow{p \mapsto e_{23}p} P_2 \xrightarrow{p \mapsto e_{12}p} P_1.$$

Determine the degree of each map and of the composite.

iii. For every $i, j \in \{1, 2, 3\}$ and $r, s \in \mathbb{Z}$, determine when there is a non-zero degree-preserving map $P_i(r) \to P_j(s)$.

Hint. An A-linear map $u : P_i \to P_j$ is determined by $u(e_{ii})$, and $u(e_{ii}) = u(e_{ii})e_{ii}$.

Exercise 2. Dependence on the symmetrising form
Let $A = \Bbbk A_2$, $S = \Bbbk e_{11} \oplus \Bbbk e_{22}$, $V = \Bbbk e_{12}$, and $M = e_{11}A$, with their multiplication actions and inherited gradings. For $c_1, c_2 \in \Bbbk \setminus \{0\}$, define

$$\tau : S \longrightarrow \Bbbk, \qquad ae_{11} + be_{22} \longmapsto c_1 a + c_2 b.$$

Recall that $M^{(*)} = \bigoplus_{d\in\mathbb{Z}} \operatorname{Hom}_S(M_{-d}, S)$, with the displayed summand in degree d. A form is symmetrising when $\tau(st) = \tau(ts)$ for all $s, t \in S$ and the pairing $(s, t) \mapsto \tau(st)$ is non-degenerate. Parts (3)–(4) concern the transported A-actions and are advanced material. For these parts, assume $\operatorname{char} \Bbbk \neq 2$.

(1) Verify that τ is symmetrising. Define $\varepsilon : V \to \Bbbk$ by $\varepsilon(ue_{12}) = u$ for $u \in \Bbbk$. Find the right S-linear map $f : V \to S$ and the left S-linear map $g : V \to S$ satisfying $\tau \circ f = \varepsilon = \tau \circ g$. Verify the stated linearities and determine whether f is also left S-linear and whether g is also right S-linear.

(2) Consider the right S-linear maps

$$\varphi_0 : M \longrightarrow S, \quad ae_{11} + be_{12} \longmapsto ae_{11},$$
$$\varphi_1 : M \longrightarrow S, \quad ae_{11} + be_{12} \longmapsto be_{22} \qquad (a, b \in \Bbbk).$$

Verify that φ_0, φ_1 form a homogeneous basis of $M^{(*)}$, and determine their degrees.

(3) Set $\tau_1(ae_{11} + be_{22}) = a + b$ and $\tau_2(ae_{11} + be_{22}) = a + 2b$. For $i \in \{1, 2\}$, define

$$\alpha_i : M^{(*)} \longrightarrow \mathbb{D}(M), \quad f \longmapsto \tau_i \circ f, \qquad a \cdot_i f = \alpha_i^{-1}(a\alpha_i(f)),$$

where $(a\lambda)(m) = \lambda(ma)$ for $a \in A$, $\lambda \in \mathbb{D}(M)$, and $m \in M$. Calculate the actions of e_{11}, e_{22}, e_{12} on φ_0, φ_1 for both forms. Are the two actions equal?

(4) Construct a graded left A-module isomorphism

$$t : (M^{(*)}, \cdot_1) \longrightarrow (M^{(*)}, \cdot_2)$$

by specifying $t(\varphi_0)$ and $t(\varphi_1)$.

Lecture 2: Tensor and quadratic algebras

Exercise 3. Interaction of tensor product with S-dual
Let S be a finite-dimensional semisimple $\Bbbk$-algebra, and let V, W be finite-dimensional S-bimodules. For any such bimodule U, let

$$U^* = \mathrm{Hom}_S(U, S)$$

be the space of right S-linear maps, with bimodule action

$$(sft)(u) = sf(tu) \qquad (s,t \in S,\ f \in U^*,\ u \in U).$$

For $f \in V^*$, $g \in W^*$, $v \in V$, and $w \in W$, consider

$$\theta_r^{V,W} : W^* \otimes_S V^* \longrightarrow (V \otimes_S W)^*,$$
$$\theta_r^{V,W}(g \otimes f)(v \otimes w) = g(f(v)w).$$

Prove that $\theta_r^{V,W}$ is a natural isomorphism of S-bimodules.

(1) Prove that the formula defines an S-bimodule map. Explicitly, check bilinearity in (g, f) and in (v, w), and show that neither of the substitutions

$$vs \otimes w \ \longleftrightarrow\ v \otimes sw, \qquad gs \otimes f \ \longleftrightarrow\ g \otimes sf \qquad (s \in S)$$

changes the value given by the formula. Check that the resulting map $V \otimes_S W \to S$ is right S-linear, and verify

$$\theta_r^{V,W}(sg \otimes ft) = s\,\theta_r^{V,W}(g \otimes f)t \qquad (s,t \in S).$$

(2) Prove that $\theta_r^{V,W}$ is bijective.

(3) For finite-dimensional S-bimodules V', W' and bimodule maps $p : V \to V'$ and $q : W \to W'$, prove

$$(p \otimes q)^* \circ \theta_r^{V',W'} = \theta_r^{V,W} \circ (q^* \otimes p^*).$$

Here $p^* : (V')^* \to V^*$ sends f to $f \circ p$; $q^* : (W')^* \to W^*$ and $(p \otimes q)^* : (V' \otimes_S W')^* \to (V \otimes_S W)^*$ are defined in the same way by composition with q and $p \otimes q$, respectively.

Hint for (2). First take $V = S$. Then use the fact that every finite-dimensional right S-module is a direct summand of $S^{\oplus n}$ for some $n \geq 0$.

Remark. The analogous statement holds for left S-duals, with left and right interchanged.

Exercise 4. Quadratic presentations and Hilbert series
When $A_0 = \bigoplus_{i=1}^{n} \Bbbk e_i$ with coordinate idempotents,

$$h_A(t)_{ij} = \sum_{d\geq 0} [e_i A_d : e_j A_0] t^d = \sum_{d\geq 0} \dim_{\Bbbk}(e_i A_d e_j) t^d \quad \text{for all } 1 \leq i, j \leq n.$$

Use the listed order of the idempotents. When $A_0 = \Bbbk$, there is just one entry.

(1) Let $A = \Bbbk A_4/(e_{13}, e_{24})$ with the quotient grading. Find a homogeneous basis, compute $h_A(t)$, and write $A = T_S(V)/(R)$ with explicit S, V, and $R \subset V \otimes_S V$.

(2) Equality of Hilbert series detects a tensor algebra when the degree 1 generators are fixed.

i. Let A be a locally finite-dimensional graded algebra generated in degree 1 over the finite-dimensional semisimple algebra $S = A_0$. Fix a bimodule ${}_S V_S$ and an S-bimodule isomorphism $\varphi : V \xrightarrow{\sim} A_1$. Prove that

$$h_A(t) = h_{T_S(V)}(t) \quad \Longrightarrow \quad T_S(V) \xrightarrow{\sim} A,$$

where the isomorphism is induced by id_S and φ and fixes S.

ii. Show that the degree 1 generation hypothesis cannot be omitted: construct a graded algebra B with $B_0 = \Bbbk$, $\dim_{\Bbbk} B_1 = 1$, and a generator in degree 2, such that

$$h_B(t) = h_{T_{\Bbbk}(B_1)}(t) \qquad \text{but} \qquad B \ncong T_{\Bbbk}(B_1)$$

as graded algebras.

Hint. For (a), use the tensor-presentation corollary and count composition factors of its kernel in each degree. For (b), consider $\Bbbk[x, z]/(x^2)$ with $\deg x = 1$ and $\deg z = 2$.

(3) Let $S = \Bbbk e_1 \oplus \Bbbk e_2$ with coordinate idempotents, and let $V = \Bbbk a \oplus \Bbbk b$ satisfy

$$e_1 a e_2 = a, \qquad e_2 b e_1 = b,$$

with all other idempotent actions zero. Place V in degree 1. Find a basis of $T_S(V)_d$ for every $d \geq 0$, and show that

$$h_{T_S(V)}(t) = \frac{1}{1-t^2} \begin{pmatrix} 1 & t \\ t & 1 \end{pmatrix}.$$

Find a homogeneous basis and the Hilbert series of $T_S(V)/(ab)$, and compare the two algebras.
Hint. Count alternating words separately in each degree, including degree 0.

(4) Let V and W have bases $x_1, \ldots, x_n$ and $y_1, \ldots, y_m$, respectively. Give every generator degree 1 and set

$$\mathrm{Sym}(V) = T_{\Bbbk}(V)/(x_i x_j - x_j x_i \mid 1 \leq i < j \leq n),$$
$$\bigwedge W = T_{\Bbbk}(W)/(w^2 \mid w \in W).$$

Prove that

$$\dim_{\Bbbk} \mathrm{Sym}(V)_2 = \frac{n(n+1)}{2}, \qquad \dim_{\Bbbk}(\bigwedge W)_2 = \frac{m(m-1)}{2}.$$

Lecture 3: The quadratic dual

For $A = T_S(V)/(R)$, use the right quadratic dual $A^! = T_S(V^*)/(R^\perp)$, with $V^* = \mathrm{Hom}_S(V, S)$ placed in degree 1. The reversed pairing is

$$\theta_r(g \otimes f)(v \otimes w) = g(f(v)w).$$

When $S = \bigoplus_i \Bbbk e_i$ and $v \in e_i V e_j$ is a chosen basis vector, its dual $v^* : V \to S$ takes v to e_j and the other basis vectors to zero; it satisfies $e_j v^* e_i = v^*$.

Exercise 5. Computing quadratic duals
The following parts are independent.

(1) **Reverse generators and complement relations.** Let $A = \Bbbk A_4/(e_{13})$. Write a presentation of $A^!$ using reversed generators, and find homogeneous bases of A and $A^!$. Check directly that dualising twice recovers A, identifying each twice reversed generator with its original matrix unit.
Hint. Among $e_{12}e_{23}$ and $e_{23}e_{34}$, only the first is a relation. Complement the forbidden composable pairs and reverse their order.

(2) **All products of two generators vanish.** Let $S = \Bbbk e_1 \oplus \Bbbk e_2$ and $V = \Bbbk a \oplus \Bbbk b$, with $e_1 a e_2 = a$ and $e_2 b e_1 = b$ and all other idempotent actions zero. Compute the quadratic dual of $T_S(V)/(ab, ba)$ and the Hilbert series of both algebras.
Check. In the order e_1, e_2, the two series are

$$\begin{pmatrix} 1 & t \\ t & 1 \end{pmatrix} \qquad \text{and} \qquad \frac{1}{1-t^2}\begin{pmatrix} 1 & t \\ t & 1 \end{pmatrix}.$$

(3) **When does every linear combination square to zero?** Write $\Bbbk\langle x, y\rangle := T_{\Bbbk}(\Bbbk x \oplus \Bbbk y)$; its basis consists of the finite words in x, y, including the empty word 1, and multiplication joins words in order. For $q \in \Bbbk$, let

$$C_q = \Bbbk\langle x, y\rangle/(x^2, y^2, xy + qyx), \qquad \deg x = \deg y = 1.$$

i. Find a homogeneous basis of C_q, compute every product of two basis elements, and determine $h_{C_q}(t)$. Treat $q = 0$ as well as $q \neq 0$, and justify that your proposed basis is linearly independent.

ii. For $u = \alpha x + \beta y \in (C_q)_1$, compute u^2. Determine, for each q, exactly which pairs $(\alpha, \beta) \in \Bbbk^2$ give $u^2 = 0$. For which q does every element of $(C_q)_1$ square to zero?

iii. Compute a quadratic presentation of $C_q^!$ using the reversed pairing.
Advanced comparison. Identify C_1 and C_{-1} as the graded and ordinary tensor products, respectively, of two copies of $\Bbbk[z]/(z^2)$ with $\deg z = 1$. When the characteristic of $\Bbbk$ is not 2, are C_1 and C_{-1} isomorphic as algebras? Explain what changes in characteristic 2.

Hint. In a candidate basis, use the class of yx for degree 2. For the dual relation, evaluate a general linear combination of X^2, XY, YX, Y^2 on $x^2, y^2, xy + qyx$, where $X = x^*$ and $Y = y^*$.

Exercise 6. Bases, Hilbert series, and quadratic duals
The following parts are independent.

(1) **A square with a parameter.** Let $S = \Bbbk e_1 \oplus \Bbbk e_2 \oplus \Bbbk e_3 \oplus \Bbbk e_4$ and let V have basis a, b, c, d, with

$$e_1 a e_2 = a, \qquad e_2 b e_4 = b, \qquad e_1 c e_3 = c, \qquad e_3 d e_4 = d,$$

and all other idempotent actions zero. The only composable words of length 2 are ab and cd. For each $\lambda \in \Bbbk$, set

$$A_\lambda = T_S(V)/(ab - \lambda cd).$$

For every $\lambda \in \Bbbk$, including $\lambda = 0$, compute a presentation of $A_\lambda^!$. Find homogeneous bases of A_λ and $A_\lambda^!$, and compute both Hilbert series. Do the series depend on λ? Recover the original relation by dualising again. For $\lambda \neq 0$, rescale d to compare A_λ with A_1.

Hint. Evaluate $\mu b^* a^* + \nu d^* c^*$ on $ab - \lambda cd$, where $\mu, \nu \in \Bbbk$.

(2) **Counting words.** Let $V = \Bbbk x \oplus \Bbbk y$ and

$$B = T_\Bbbk(V)/(x^2), \qquad \deg x = \deg y = 1.$$

i. Describe a basis of B_d for every $d \geq 0$, and compute $a_d := \dim_\Bbbk B_d$ for $0 \leq d \leq 6$.

ii. For $d \geq 2$, find an equation expressing a_d in terms of a_{d-1} and a_{d-2} by sorting the basis words according to their first letter. Use the equation and the values of a_0, a_1 to find $h_B(t)$ as a rational function.

iii. Write $X = x^*$ and $Y = y^*$. Compute $B^!$, find a homogeneous basis, and determine $h_{B^!}(t)$. Which of B and $B^!$ are finite-dimensional?